

9. Request and Response Validation

9.1 Overview of Request and Response Validation

Request and Response Validation is one of the most important activities in API testing. The purpose of validation is to ensure that the API accepts requests correctly and returns accurate responses according to the business requirements and API documentation.

In this project, every API request sent to the ReqRes server is validated by checking:

- HTTP Status Codes
- Response Body Content
- Response Time Performance
- JSON Schema Structure

These validations help ensure that the API behaves correctly, consistently, and reliably.

9.2 Status Code Validation

What is Status Code Validation?

When a client sends a request to an API, the server returns an HTTP status code indicating the result of the request.

Status code validation ensures that the API returns the correct HTTP response code based on the operation performed.

Why Status Code Validation is Important

Status codes provide the first indication of whether an API request was successful or failed.

They help testers:

- Verify API functionality
 - Detect failures quickly
 - Validate error handling
 - Ensure compliance with API standards
-

Status Codes Used in This Project

200 OK

Indicates that the request was successfully processed.

Used In:

- Login API
- Get Users API
- Get Single User API
- PUT Update API
- PATCH Update API

Example

GET /users/2

Response:

200 OK

201 Created

Indicates that a new resource has been successfully created.

Used In:

- Create User API

Example

POST /users

Response:

201 Created

204 No Content

Indicates that the request was successful and there is no content to return.

Used In:

- Delete User API

Example

DELETE /users/2

Response:

204 No Content

404 Not Found

Indicates that the requested resource does not exist.

Used In:

- Invalid User API

Example

GET /users/23

Response:

404 Not Found

Postman Test Script Example

```
pm.test("Status Code Validation", function () {  
    pm.response.to.have.status(200);  
});
```

Expected Outcome

The API should return the correct status code for each operation according to the API documentation.

9.3 Response Body Validation

What is Response Body Validation?

Response Body Validation verifies that the data returned by the API is accurate, complete, and matches expected values.

The response body is typically returned in JSON format and contains the actual data requested by the client.

Why Response Body Validation is Important

A status code alone cannot guarantee that the returned data is correct.

For example:

- Status Code may be 200
- But data may be incorrect or incomplete

Therefore response content must also be validated.

Login API Validation Example

Request:

```
{
  "email": "eve.holt@reqres.in",
  "password": "cityslicka"
}
```

Response:

```
{
  "token": "QpwL5tke4Pnpja7X4"
}
```

Validation Performed

- Token exists
- Token is not empty

Postman Script

```
pm.test("Token Exists", function () {
  var response = pm.response.json();
  pm.expect(response.token).to.exist;
});
```

Get Single User Validation Example

Response:

```
{
  "data": {
    "id": 2,
    "email": "janet.weaver@reqres.in",
    "first_name": "Janet",
    "last_name": "Weaver"
  }
}
```

```
}  
}
```

Validation Performed

- User ID exists
- Email exists
- First name exists
- Last name exists

Postman Script

```
pm.test("User Data Validation", function () {  
    var jsonData = pm.response.json();  
  
    pm.expect(jsonData.data.id).to.eql(2);  
    pm.expect(jsonData.data.email).to.exist;  
    pm.expect(jsonData.data.first_name).to.exist;  
});
```

Expected Outcome

The API should return accurate and complete data according to the request made by the user.

9.4 Response Time Validation

What is Response Time Validation?

Response Time Validation measures how quickly the API responds after receiving a request.

It helps identify performance issues and ensures users receive data within an acceptable time.

Why Response Time Validation is Important

Slow APIs can:

- Affect user experience
- Increase application load times
- Cause system bottlenecks

Therefore response performance must be monitored.

Response Time Measurement

Postman automatically displays response time in milliseconds (ms).

Example:

Response Time: 145 ms

Validation Criteria

For this project:

Response Time	Result
Less than 1000 ms	Pass
Greater than 1000 ms	Review Required

Postman Script Example

```
pm.test("Response Time Under 1000 ms", function () {  
    pm.expect(pm.response.responseTime).to.be.below(1000);  
});
```

Expected Outcome

All ReqRes APIs should respond within the acceptable performance threshold.

9.5 JSON Schema Verification

What is JSON Schema Verification?

JSON Schema Verification ensures that the API response follows the predefined structure and data format.

Instead of checking individual fields manually, schema validation verifies the entire response structure.

Why JSON Schema Verification is Important

It helps:

- Detect missing fields
 - Detect incorrect data types
 - Ensure consistency across responses
 - Improve automation reliability
-

Example Response

```
{
  "data": {
    "id": 2,
    "email": "janet.weaver@reqres.in",
    "first_name": "Janet",
    "last_name": "Weaver"
  }
}
```

Example JSON Schema

```
{
  "type": "object",
  "properties": {
    "data": {
      "type": "object",
      "properties": {
        "id": {
          "type": "number"
        },
        "email": {
          "type": "string"
        },
        "first_name": {
          "type": "string"
        },
        "last_name": {
          "type": "string"
        }
      },
      "required": [
        "id",
        "email",
        "first_name",
        "last_name"
      ]
    }
  }
}
```

Postman Schema Validation Script

```
const schema = {
  "type": "object",
```

```
    "properties": {
      "data": {
        "type": "object",
        "properties": {
          "id": {"type": "number"},
          "email": {"type": "string"},
          "first_name": {"type": "string"},
          "last_name": {"type": "string"}
        },
        "required": [
          "id",
          "email",
          "first_name",
          "last_name"
        ]
      }
    }
  };

pm.test("Schema Validation", function () {
  pm.response.to.have.jsonSchema(schema);
});
```

9.6 Validation Summary

Validation Type	Purpose
Status Code Validation	Verify request success or failure
Response Body Validation	Verify returned data accuracy
Response Time Validation	Verify API performance
JSON Schema Verification	Verify response structure consistency

9.7 Final Conclusion

Request and Response Validation is a critical part of API testing because it confirms that the API is functioning correctly, returning accurate data, maintaining performance standards, and following the expected JSON structure.

In this project, validation is performed using Postman test scripts for every API endpoint, ensuring comprehensive quality checks across authentication, retrieval, creation, update, and deletion operations. This approach closely reflects the API validation practices used in professional QA and software testing environments.